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Measurement approaches for greenhouse gas emissions from rice I: technical evolution and scientific results obtained with different methods

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Rice fields are a critical source of CH₄ and N₂O, necessitating accurate, field-level measurements to inform effective mitigation programs. This review (Volume 1) offers a novel perspective on this topic by focusing on the measurement systems themselves, driven by one core question: To what extent have the technical capabilities and limitations of field measurement systems shaped the current scientific understanding and knowledge gaps on rice greenhouse gas (GHG) emissions? We provide a comprehensive assessment of three major field approaches: Manually Sampled Chambers (MSC), Fully Automated Chambers (FAC), and Eddy Covariance (EC). Reversing the narrative of typical literature reviews that focus primarily on scientific findings, this paper starts with the technical evolution of each method, followed by a comparative assessment based on frameworks for method selection and scientific key contributions. The current scientific consensus and global estimates are overwhelmingly derived from the highly versatile MSC approach, which has generated a vast database across different rice-growing regions and management treatments, enabling statistically robust meta-analyses. Despite limitations—such as altering the microclimate in the headspace, possibly missing diurnal or seasonal peaks, and limited spatial scalability—the MSC remains a cornerstone of rice GHG research and will continue to play a central role. The FAC system was developed as an alternative, overcoming limitations in sampling frequency and providing robust data on diurnal and seasonal emission patterns, which proved especially valuable in comparative studies on crop management impacts. Finally, we discuss the use of EC, which provides high-resolution, integrative datasets that allow for a greatly improved process-based understanding of GHG fluxes. The established FluxNet collaboration of EC researchers could serve as a blueprint to coordinate chamber-based studies, thereby building the comprehensive dataset necessary to support data-driven modeling and Machine Learning (ML) development.

This retrospective assessment in Volume 1 establishes a critical framework for evaluating and selecting rice GHG measurement methods. Volume 2 of this paper will supplement this work by addressing emerging technical innovations and prospects against the backdrop of diversified research objectives.

KEYWORDS

climate change mitigation, closed chamber technique, direct sampling, eddy covariance, gas chromatography, manual sampling, methane, nitrous oxide

1 Introduction

Rice is a staple food for much of the world's population and its production accounts globally for approximately 10–11% of agricultural methane emissions (IPCC (Intergovernmental Panel on Climate Change), 2022). Rice cultivation practices contribute significantly to greenhouse gas (GHG) emissions, particularly methane (CH₄), a short-lived GHG with a global warming potential (GWP) of 27 over a 100-year horizon and 78 over a 20-year horizon according to the latest IPCC Report (Masson-Delmotte et al., 2021). Rice paddies are a source of nitrous oxide (N₂O), which is a by-product of fertilizer application to the soil. Although this long-lived GHG has a high GWP of 273 under both 20- and 100-year time horizons (Masson-Delmotte et al., 2021), the quantities of N₂O emitted from rice fields are relatively low as compared to CH₄, so its overall contribution to the GHG budget of rice is typically less than 10% (Qian et al., 2023). In many Asian countries, the relative share of rice production is more than 50% of agricultural emissions as documented in GHG inventories within the respective National Communications¹. These national quantifications are calculated following the IPCC Guidelines, requiring emission factors (EF) and activity data. As for the latter, the datasets comprise the rice area under a given crop management, which can be obtained – in the best case – from national statistics, or have to be estimated based on expert knowledge. The initial GHG inventories in the 1990s were based on default EFs from the IPCC Guidelines (Tier 1). However, the IPCC encourages national institutions to conduct field measurements as a means to determine country-specific EFs for Tier 2 estimates. In recent years, many rice-growing countries initiated their measurement campaigns to determine country-specific EFs. Despite a growing number of published field measurements, the available databases on national and subnational EFs are still small against the total number of different rice ecosystems under varying natural and agronomic settings (Vo et al., 2020). These apparent shortcomings in the coverage of EFs in all rice-growing countries demand more field measurements, which sets out the context and motivation for this study.

Methane is produced in waterlogged fields, which is a unique feature of rice production in contrast to other cereals. The finding on significant CH₄ emissions from rice paddies was initially derived

from incubation studies (Koyama, 1963) and was then integrated into the assessments of the global methane budget (Ehhalt, 1974). The quantification of *in situ* flux rates was initially deduced from emission measurements in natural wetlands that were considered analogs for rice fields (Baker-Blocker et al., 1977). Several studies in the 1970s investigated the significance of microbial CH₄ production and the storage of CH₄ in soils, also reported on the release of this gas to the atmosphere (Yoshida et al., 1975; Araragi and Tangcham, 1979). The intended reduction in the GHG emissions from rice fields will rely on profound knowledge of their variation over space and time as affected by the environment and agronomic practices. Given the exponentially growing publications on GHG emissions from rice production (Yuan and Sun, 2023), several reviews and meta-analyses on GHG emissions from rice were published recently (e.g. Qian et al., 2023; Nikolaisen et al., 2023) that were almost exclusively based on the manually sampled chambers (MSC) approach. Here, we provide a genuinely new perspective on the current state of knowledge, namely by considering the scientific contributions and technical limitations of the MSC approach vis-à-vis field measurements deploying the fully automatic chambers (FAC) and Eddy Covariance (EC) techniques.

Our review was motivated by one underlying question: To what extent did the technical capabilities and limitations of field measurement systems shape the current state of knowledge on GHG emissions from rice –alongside the existing knowledge gaps? In line with this objective, the narrative starts with the technical evolution of each approach, followed by a comparative assessment based on two frameworks for method selection and scientific key contributions, respectively. Collectively, this perspective reverses the narrative of other literature reviews that focus on the findings and, to a lesser extent, on the underlying field technologies. This retrospective assessment of measurements in Volume 1 is supplemented by Volume 2 of this paper (Vo et al., 2026), which deals with emerging technical innovations and prospects against the backdrop of diversified research objectives and project settings for GHG emissions from rice. In Volume 2, we explore the potential of laser-based trace gas analyzers for rapid “fast-box” measurements that are tested in a synchronized field experiment. Given the specific purpose of Volume 1 for providing a retrospective assessment of measurement approaches, these advanced analytical approaches as well as the potential benefits stemming from Machine Learning

¹ <https://unfccc.int/non-annex-I-NCs>

(ML) and Remote Sensing (RS) are here only addressed when needed for setting a broader context.

2 Manually sampled chambers

2.1 Technical evolution

Early attempts to measure GHG emissions using the closed static chamber technique were developed for upland systems and particularly for measuring N₂O emissions from soils by Freney et al. (1978) and Hutchinson and Mosier (1979). In rice fields, however, the measurement of emissions, particularly for CH₄, requires the enclosure of entire rice plants, considering that the plant acts as a conduit for the transport/emission of this GHG to the atmosphere. This led to the use of chambers that were high enough to reduce disturbance and risks of mechanical damage to the plant canopy. The first known emission measurement in rice fields was done using a “carboy” attached to a floating collar adapted for flooded fields (Cicerone and Shetter, 1981). Several research groups tested distinct designs of static chambers, including a chamber hanging on an external frame (Seiler et al., 1983), before the current type of chamber was widely used and considered standard for emission measurements in rice fields (Holzapfel-Pschorn and Seiler, 1986).

Because of the versatility and simple infrastructural requirements, closed chamber measurements were used in numerous field experiments in almost all rice-growing countries, including early adoption of this approach in several Asian countries: China (Khalil et al., 1990; Wassmann et al., 1993a, b), India (Adhya et al., 1994; Parashar et al., 1994), Japan (Kimura, 1992), Thailand (Kimura, 1992), and the Philippines (Van der Gon et al., 1992; Wassmann et al., 1994). In these studies, the top chamber or collector box is placed on a basal metal frame positioned on the soil surface. The air inside the headspace is mixed by a fan, and samples are retrieved using a syringe and then stored in vials. In terms of sampling, some published field studies used a pump for sample transfer (e.g., Khalil and Rasmussen, 1983). Gas was stored in stainless steel flasks. The use of syringes and vials has become the most common approach over the years.

Despite the common features in the prevailing prototypes outlined above, the chambers used by different researchers come in various shapes and sizes (Sander and Wassmann, 2014). Chambers can be either transparent or opaque. The use of opaque chambers is generally justified by the fact that CH₄ is transported through rice plants via aerenchyma, and not through stomata. In turn, direct light-induced stomatal activity doesn't control methane flux. On the other hand, the exclusion of sunlight will ultimately affect physiological processes and transparent material was used as chamber walls in most studies. Nevertheless, opaque chambers are generally also considered suitable tools for GHG emission measurements in rice fields derived from publications addressing the bio-physical aspects (Zaman et al., 2021) as well as the practicality of field

measurements at multiple locations (Minamikawa et al., 2015). A basal frame is fixed on the soil surface throughout the cropping season to minimize soil disturbance when placing the chambers. The connection between the basal frame and the chamber has to be airtight, which is typically achieved through a horizontal trough filled with water (Minamikawa et al., 2015).

The installation of a small vent tube allows for pressure equalization between the chamber headspace and ambient air, thereby reducing pressure-related artifacts during chamber deployment (Zaman et al., 2021). While the vent minimizes the micro-meteorological impacts of the chamber, the possible dilution may also affect the temporal concentration gradient. While published data support the use of a pressure release specifically after chamber placement (Clough et al., 2020), the use of non-vented static chambers remains common practice, and pressure effects can also be minimized through appropriate chamber design and deployment times. It should be noted that the vented chamber type should not be mixed up with dynamic chambers that have a defined inflow and outflow. The flux calculations based on the concentrations in inflowing and outflowing air are required for reactive gases, namely NO, NO₂, and NH₃ (Zaman et al., 2021), but this chamber type is outside of the scope of this review, focusing on closed chambers for CH₄ and N₂O emissions. As for the analytical procedure, the vast majority of chamber measurements were conducted in combination with GC analysis, but in some studies, sensors were used instead of a GC (Yao et al., 2013). Photo-acoustic sensors for CH₄/N₂O detection have been commercially available for several decades but have only rarely been used in GHG measurements in rice fields. These instruments require fairly elaborate and frequent calibration procedures (Tirol-Padre et al., 2014), which may – in addition to high investment costs – be one of the reasons for the limited adoption of this technique. As for the next generation of GHG assessments, we see an emerging adoption of laser-based instruments, which will be discussed in detail in Volume 2 of this paper.

The diversified protocols used by different research groups demand standardization of the implementation of the MSC approach for compiling coherent data sets in emission assessments (Parkin and Venterea, 2010). Dorich et al. (2020) developed guidelines for gap filling to achieve more compatibility in the seasonally accumulated emissions. The basic features of their approach comprise (i) interpolation-based gap-filling through temporal interpolation techniques—such as linear or spline interpolation, (ii) strategic sampling design by advocating for targeted sampling during key phenological stages, (iii) standardization across studies to ensure comparability across different field studies and regions, and (iv) improved seasonal estimates by focusing on high-emission periods, even when full continuous monitoring is not feasible. On the operational level of these measurements, the accuracy might be improved by longer placements of the chambers, resulting in larger concentration gradients to be detected. During the daytime, however, the duration of the placement is confined by possible plant damage

caused by high temperatures and low CO₂ concentrations. Thus, Wassmann et al. (2018) suggested GHG measurement during night-time derived from a significant correlation between nocturnal emissions and the full 24-h periods. Another approach to deal with gaps is the recommendation of strategic data points for both diurnal and seasonal cycles to optimize the reliability of extrapolations under given labor resources (Buendia et al., 1998; Brye et al., 2017). The selection of crucial sampling dates can also be tied into crop management events, namely right after drainage periods for CH₄ and fertilizer application for N₂O, which can be derived from an MSC field experiment with daily sampling throughout the season as published by Chu et al. (2015).

2.2 Significance and applications

This synthesis of the scientific findings obtained by using the closed chamber method was written by taking into account the recently published reviews and meta-analyses of the scientific findings from published field studies on GHG emissions from rice. Although not explicitly mentioned in these publications, it should be noted that these scientific assessments were almost exclusively based on the MSC approach. To minimize redundancies, we refrain here from repeating research findings from individual studies reviewed in these papers and only summarize the global statistics that cut across the aggregated database.

- Qian et al. (2023) conducted a meta-analysis of more than 400 field studies and calculated average emission rates accounting for the entirety of the published studies, cutting across the climate and management variability. The global averages of the area-scaled and yield-scaled GHG emissions of rice are ~7,870 kg CO₂ ha⁻¹ yr⁻¹ and 0.9 kg CO₂e kg⁻¹, respectively.
 - As for the mitigation effects, Qian et al. (2023) quantified the average GHG reduction through straw removal strategies (46%), non-continuous flooding (44%) and new rice variety selection (24%). It should be noted, however, that the empirical bases for these three assessments vary over a large range. To illustrate this assumption, we assessed the studies listed in this meta-analysis, which include more than 100 assessments for water management, more than 60 for straw management, but only 14 for variety selection.
 - Based on a meta-analysis of 91 publications from China, Ma et al. (2024) concluded that a combination of water and fertilizer practices could reduce CH₄ emissions of up to 67% which is markedly higher than applying these practices independently.
 - Nikolaisen et al. (2023) assessed a database of 110 published studies and provided averages on crop management impacts, including a quantification of the pre-season water management. A long drainage reduced CH₄ emissions by 23% as compared to fields with short drainage, although the actual effect of pre-season water regime depends on the cropping system and differs for a long winter fallow.
 - The impact of organic amendments on CH₄ emissions largely depends on the nature of the incorporated material. A meta-analysis by Linquist et al. (2012) found that farmyard manure (FYM) increased CH₄ emissions by 26% and the green manure (GrM) *Sesbania* by 192% when compared to inorganic N fertilizers. A special case is given by straw amendments that are not considered organic fertilizers as such, but caused, on average, a 50% higher increase as compared to livestock manure (Qian et al., 2023).
 - In terms of N₂O emissions, Qian et al. (2023) quantified the relative impact of N application compared with a situation without N fertilizer, which resulted in an increase in N₂O emissions by 182% on average. As for the practical options for mitigation, however, it will be more meaningful to assess the percentage of applied N emitted directly as N₂O-N, as done in the IPCC Guidelines. A global analysis of published emission factors by Cui et al. (2021) resulted in an average of 0.52% (0.15–1.3%).
 - Yang et al. (2022) synthesized the results of 46 field studies and analyzed the impact of several parameters, including enhanced-efficiency nitrogen fertilizer (EENF). Overall, EENF, including inhibitors and slow-release fertilizer, significantly reduced CH₄ emission by 16%. It should be noted that this ratio does not consider off-site emissions from fertilizer production that would enhance the mitigation potential of EENF under steady yield potentials.
 - As for the global estimates – that are directly or indirectly derived from chamber measurements – the available data indicate a range of 20 to 35 Tg CH₄, although there are diverging assessments on the ongoing trends at the global scale (Qian et al., 2023). Given the steady increase in average rice yields, the global yield-scaled CH₄ emissions have decreased by 38–55% over recent decades, which can be substantiated by data at the country level.
 - Shang et al. (2021) conducted a global meta-analysis on more than 300 observation sets of non-CO₂ emissions (CH₄ and N₂O) and more than 400 of soil organic carbon (SOC). In their analysis, they stressed the significance of the SOC decline triggered by intermittent irrigation for the overall GHG budget. While this impact seems plausible in the initial years, it was not clear from their analysis how long this trade-off will be significant, given a new steady state of SOC levels evolving within a few years and a finite amount of soil carbon stocks at the start of this transition.
 - Yagi et al. (2020) conducted a meta-analysis to analyze the potential of technical mitigation options in the regional context of Southeast Asia. Alternate Wetting and Drying (AWD), a water-saving technique encompassing non-flooded periods, significantly reduced CH₄ emissions by 35% (95% confidential interval: 41–29%) as well as the combined effects of CH₄+N₂O (net GWP) by 29% (36–23%).
- In addition to these science-driven assessments, the MSC approach played a key role in GHG inventories as part of the

National Communications² that have to be updated regularly. After the initial rounds with global default emission rates (IPCC Tier 1) applied by most countries, several countries have now initiated measurement campaigns based on the MSC approach to determine national emission factors as the basis for using the IPCC Tier 2 approach for rice. This group includes major rice-growing countries in Asia such as China, India, Indonesia, Thailand, Vietnam and the Philippines. Field measurements have also been published for several other Asian countries (e.g. South Korea, Bangladesh and Sri Lanka), but those emission rates were not officially adopted in Tier 2 estimates for rice in their most recent National Communications.

The MSC approach is also the standard method for field measurements in carbon projects targeting both the Voluntary and Compliance Markets. The current methodologies for rice projects by the UNFCCC³ and Gold Standard⁴ and VERRA⁵ uniformly prescribe the MSC approach as the only option for field measurements, although it should be noted that they also allow for GHG calculations solely based on emission factors without field validations. As the carbon markets dealing with rice projects are presently still at an emerging state, it remains to be seen if field measurements will in the future be seen as an add-on asset of projects and if this data will end up in scientific publications.

3 Fully automated chambers

Building on the measurement principle of the MSC approach, a Fully Automated Chamber (FAC) system consists of an integrated setup encompassing air transfer from the chamber headspace for analysis in an adjacent field lab. This system consists of the following components⁶:

- Chambers are placed on the rice field throughout the season and equipped with a cover for automated enclosure of entire rice plants in sequential cycles, i.e. each chamber is closed for 12–20 minutes within a cycle of 2–4 hours.
- A sampling system consisting of pumps and an integrated multi-valve unit to facilitate direct air transfer from the headspace of the chamber that is closed at that time, and

- A gas chromatograph (GC) configured for CH₄ detection is equipped with automated injection of air samples and calibration gas at regular intervals.

The entire system is fine-tuned to operate autonomously on a schedule, transferring headspace gas directly to the GC without intermediate storage (e.g., vials), enabling high-frequency, *in-situ* CH₄ analysis (Butterbach-Bahl et al., 2011; Tokida, 2021). Despite similarity in the principal measurement features, the individual systems developed by different research groups show modifications in terms of the control system (installed in a PC or as a microcontroller), relay systems to control the solenoid valves, data logger, etc. Moreover, these systems are typically attached to environmental sensors, e.g., soil/water temperature. The emission rates are computed with the identical equation as for manual chambers, derived from the CH₄ accumulation rate (ppm/min), temperature, and chamber volume.

The technical development of automated systems started in 1996 with measurements in Italian rice fields (Schuetz et al., 1989). This approach was also applied through individual systems in China (Wassmann et al., 1993a), the Philippines (Wassmann et al., 1994) and Japan (Yagi et al., 1996) before its further adoption through a network of research stations for CH₄ emissions from rice fields in five Asian countries, namely the Philippines, China, India, Indonesia and Thailand (Wassmann et al., 2000a). It should be noted that this list only comprises the first publication for a given country because the number of publications on FAC studies has considerably increased in recent years. Though not explicitly addressing rice fields, the efforts on standardization of automated chambers by the European ICOS network (Pavelka et al., 2018) as well as the technical descriptions by the Global Research Alliance (Grace et al., 2020) can serve as a reference for adequate design and operational guidelines of these systems.

Several field studies encompassed a comparison of emission rates obtained with automated vs manual systems that yielded generally compatible results (Zheng et al., 1998; Yao et al., 2009; Mumu et al., 2024). Given the continuous placement of each chamber on a given plot throughout the entire season, Yu et al. (2006) observed a yield decrease inside their chambers, although other research groups found no yield impact when using transparent material, ventilated design and short enclosure periods within one sequential cycle (Tokida, 2021).

Right from the onset of automated chambers used in rice fields, special emphasis was given to diurnal and seasonal cycles (Schuetz et al., 1989; Wassmann et al., 1994; Yagi et al., 1996). The database of continuous measurements was then used to devise guidelines on how to optimize manual sampling by selective timing. Buendia et al. (1998) analyzed a data set from several stations in Asia and concluded that sampling at 0600, 1200, and 1800 h will capture the diurnal variation most efficiently. It should be noted, though, that field measurements at 0600 were not adopted by any research group due to practical implications for the technical staff. Instead, the technical guidelines recommend sampling in two possible time windows, between 0900 to 1100 or 1500 to 1700, to avoid the

2 The most recent National Communications can be obtained at <https://unfccc.int/ghg-inventories-annex-i-parties/2023> for Annex I countries and <https://unfccc.int/national-reports-from-non-annex-i-parties> for non-Annex I countries

3 <https://cdm.unfccc.int/UserManagement/FileStorage/5IP163JN4RK62D0XOQZS9T7W8MEYAC>

4 <https://globalgoals.goldstandard.org/437-luf-agr-methane-emission-reduction-awm-practice-in-rice/>

5 <https://verra.org/methodologies/improved-management-in-rice-production-systems/>

6 A simplified technical description of these systems is available at <https://ghgmitigation.irri.org/resources/guidelines/measurements-approaches/automated-chamber-systems>

diurnal peak that typically occurs around midday. Regarding the seasonal estimates, their recommendation derived from continuous records was a high sampling frequency at the final stage, whereas the sampling intervals can be expanded during the mid-season. Moreover, automated chamber systems were used to quantify emissions during fallow seasons, which closed a sizable gap in the annual CH₄ budgets (Bronson et al., 1997b).

FAC systems were also used to describe the CH₄ emission of the three rice ecosystems characterized by different types of water regimes, namely irrigated, rainfed and deepwater rice (Wassmann et al., 2000b). Though backed up by other studies using MSC, the results of this multi-country network of FAC stations were reflected in the scaling factors of these ecosystems given in the IPCC Guidelines (Ogle et al., 2019).

In the context of mitigating GHG emissions, automated systems yielded important information to substantiate the impact of different crop management practices. Several FAC studies investigated the impact of Alternate Wetting and Drying by quantifying (i) the time lag in emission records following an individual drainage event, (ii) the impact of the timing of the drainage events as a function of plant stages and (iii) the overall mitigation effect with higher accuracy than possible by only periodic manual sampling. The findings related to the adoption of AWD practices indicated a mitigation effect of suppressing CH₄ emissions up to 60% in the Philippines (Bronson et al., 1997a) and 38% in Japan (Yagi et al., 1996).

The effect of shifting from a flooded rice system to an entirely non-flooded/aerobic system was found to reduce the GWP by about 90% which can be attributed to a drastic decline in CH₄ (95%), superseding the increase of N₂O emissions (Weller et al., 2015). As the bulk of CH₄ reaching the atmosphere is emitted through the rice plants, automated systems were also used for the comparative assessments of different varieties (Wassmann et al., 2002; Setyanto et al., 2004). Although these comparisons revealed some differences, the overall picture of the impact of variety selection on CH₄ emissions remains unclear – even under consideration of the variety-screening of CH₄ transport capacities described below.

In terms of fertilization, the use of an automated system was especially instrumental in capturing the short-term spikes in emission that might be missed through manual sampling in weekly intervals. Deep placement of urea had lower N₂O emissions as compared to surface application, although the reduction effect varied from 14 – 64% depending on the season (Gaihre et al., 2015). Abao et al. (2000) tested slow-release N-fertilizer and observed a drastic decrease (97%) in N₂O emissions while CH₄ emissions remained largely unaffected. Fertilizing with ammonium sulfate instead of urea generally reduced CH₄ emissions, varying from 18 to 201% depending on the season (Corton et al., 2000), although other studies showed an inconsistent impact (Bronson et al., 1997a).

FAC measurements on the impact of different types of organic amendments (Yagi et al., 1996; Wassmann et al., 1996; Adhya et al., 2000) supplemented the database on incremental CH₄ emissions obtained with manual chambers and also led to several new insights into the underlying mechanisms. The importance of pre-

fermentation on CH₄ emissions was highlighted through a comparative field study in which the soil incorporation of biogas slurry reduced CH₄ emissions by 61% as compared to fresh organic material based on an identical N amount (Wassmann et al., 1993b). Moreover, the FAC systems were also used for N₂O measurements (Abao et al., 2000; Weller et al., 2015; Janz et al., 2019; Gaihre et al., 2015). As long as those systems are optimized for CH₄, the required chamber height inherently constrains detection limits of GHG emissions, resulting in near-horizontal concentration gradients in the case of low emissions (see sub-chapter on detecting low N₂O emissions). On the other hand, the continuous nature of the FAC records encompasses the coverage of those episodes with moderate and higher emission rates that allow reliable slope detections. In turn, the resulting cumulative data set typically suffices for quantifying N₂O emissions at the seasonal and annual scales with reasonable accuracy.

Automated chamber systems were also deployed for process-based studies assessing the impact of soil temperature on CH₄ emissions (Schuetz et al., 1990) or different transport mechanisms of CH₄. The plant-mediated transport of different rice cultivars was quantified by developing a specific plant chamber that followed a similar principle as the automated field chambers (Butterbach-Bahl et al., 1997; Aulakh et al., 2000). The continuous records of automated field systems allowed a distinction of CH₄ emitted in the form of bubbles during the season (Tokida, 2021), and entrapped CH₄ from wetland rice fields upon soil drying (Van Der Gon et al., 1996). Using automated chambers installed between the plant rows, Komiya et al. (2015) found that CH₄ bubble ebullition was correlated with falling atmospheric pressure.

Several automated systems were operated alongside other studies investigating different aspects of the rice ecosystem. The emission data were then used as background information to set the context for other research findings, e.g., the impact of low-emission rice cultivation on the soil fauna (John et al., 2020). The continuous data obtained with these systems proved especially valuable for validating process-based models, e.g. the Landscape-DNDC model developed in parallel with a large field experiment in the Philippines (Kraus et al., 2015).

4 Eddy covariance

Eddy covariance is a micrometeorological technique used to quantify the atmospheric exchange of gases and water vapor from a given ecosystem. This method relies on high-frequency measurements (typically 10 Hz or higher) of the 3-dimensional wind field and the gas concentrations (Baldocchi, 1997). Recording wind speed and direction can then be used to calculate the upward and downward flows of these gases – resulting in the net emissions over a specific area. As the application of EC relies on fast-response gas sensors and powerful data acquisition systems, the number of studies took a marked increase when prices of these instruments declined in the 1970s.

However, the initial decades of EC studies were confined to recording heat, water vapor and CO₂ exchange. As such, EC evolved

into a widely used tool to describe ecosystem functions at high temporal resolution. EC was frequently used to determine the net balance of atmospheric CO₂ exchange as a proxy for determining carbon sequestration or release of a given ecosystem. In this context, the EC technique was key to large-scale programs studying the role of forests in the global carbon budget.

The EC measurements of non-CO₂ GHGs, namely CH₄ and N₂O, presented challenges due to their low atmospheric concentrations and the need for fast-response, high-precision gas analyzers. The high-resolution equipment based on tunable diode laser absorption spectroscopy (TDLAS) became commercially available around the year 2000, enabling its integration into eddy covariance systems. As for CH₄, the initial EC measurements with this instrumentation were done in natural peatlands (Hendriks et al., 2007) and those with agricultural use (Schrier-Uijl et al., 2010) as well as in landfills (Eugster and Plüss, 2010). At the same time, initial EC measurements of N₂O were based on different laser technologies (Eugster et al., 2007) to assess fluxes from intensive livestock production.

In the specific case of rice fields, the pioneering micrometeorological measurements of CH₄ fluxes were conducted by using the flux-gradient method to determine CH₄ fluxes in the Philippines (Simpson et al., 1995) and Japan (Miyata et al., 2000). While this technique is an indirect approach and relies on the vertical gradient in concentrations, the EC method has, in the meantime, evolved as the standard micrometeorological technique for rice fields and other ecosystems with a homogenous terrain. Numerous studies have applied EC for CH₄ measurements in rice fields, including those conducted in the USA (Detto et al., 2011; Runkle et al., 2018), Taiwan (Tseng et al., 2010), Italy (Werle and Kormann, 2001), Philippines (Alberto et al., 2014), Odisha/India (Swain et al., 2018), Jiangsu Prov./China (Ge et al., 2018), Korea (Kang et al., 2015), and Thailand (Kanyatrakul et al., 2023). As previously done for FAC, our list only comprises the first publication of a given region or country because the number of publications on CH₄/EC studies has considerably increased in recent years. EC measurements in rice require a “fetch” of 100 m radius, translating into one 4-ha plot with a homogeneous treatment (Reba et al., 2020). In turn, almost all EC measurements in rice fields only assessed a single treatment except for those using a “roving tower” alternating between two different fields (Alberto et al., 2014).

Several EC studies made their data publicly available on the FLUXNET-CH₄ database (<https://fluxnet.org/data/>). This global database comprises aggregated CH₄ fluxes and meteorological data from 79 sites, including 7 sites with rice fields located in the US (2x Arkansas, 1x California), Korea, Japan, China, Italy and the Philippines (Delwiche et al., 2021). It should be noted, though, that the number of sites in the tropics is generally rather limited in FLUXNET-CH₄, which is also reflected for rice fields with only one tropical site. These FLUXNET rice sites have average annual emissions across all sites of 16.7 ± 7.7 g CH₄-C m⁻² yr⁻¹ and are characterized by strong seasonal patterns, with either one or more CH₄ emission peaks per year depending on the number of rice seasons and field water management (Delwiche et al., 2021). Flooded rice fields are typically associated with one peak during

the reproductive period of each season, which may be accompanied by a secondary peak due to drainage events (Alberto et al., 2014; Runkle et al., 2018).

Independent from the FLUXNET database, Ouyang et al. (2023) combined paddy-rice methane-flux data from 23 global eddy covariance sites and MODIS remote sensing data with ML to predict rice CH₄ fluxes. Their gridded up-scaling estimates for rice across Monsoon Asia, covering almost 90% of the global rice area, accounted for 20.6 ± 1.1 Tg CH₄ yr⁻¹ corresponding to about 78.6 g CH₄ m⁻² yr⁻¹, which was at the lower range of previous inventory-based estimates. They also concluded a declining trend in emissions starting around 2007, which excludes that rice has contributed to the rapid increase in CH₄ concentrations observed in recent years (Ouyang et al., 2023). These two recent studies based on global networks demonstrate the potential of EC data for clarifying fundamental questions of the atmospheric CH₄ budget.

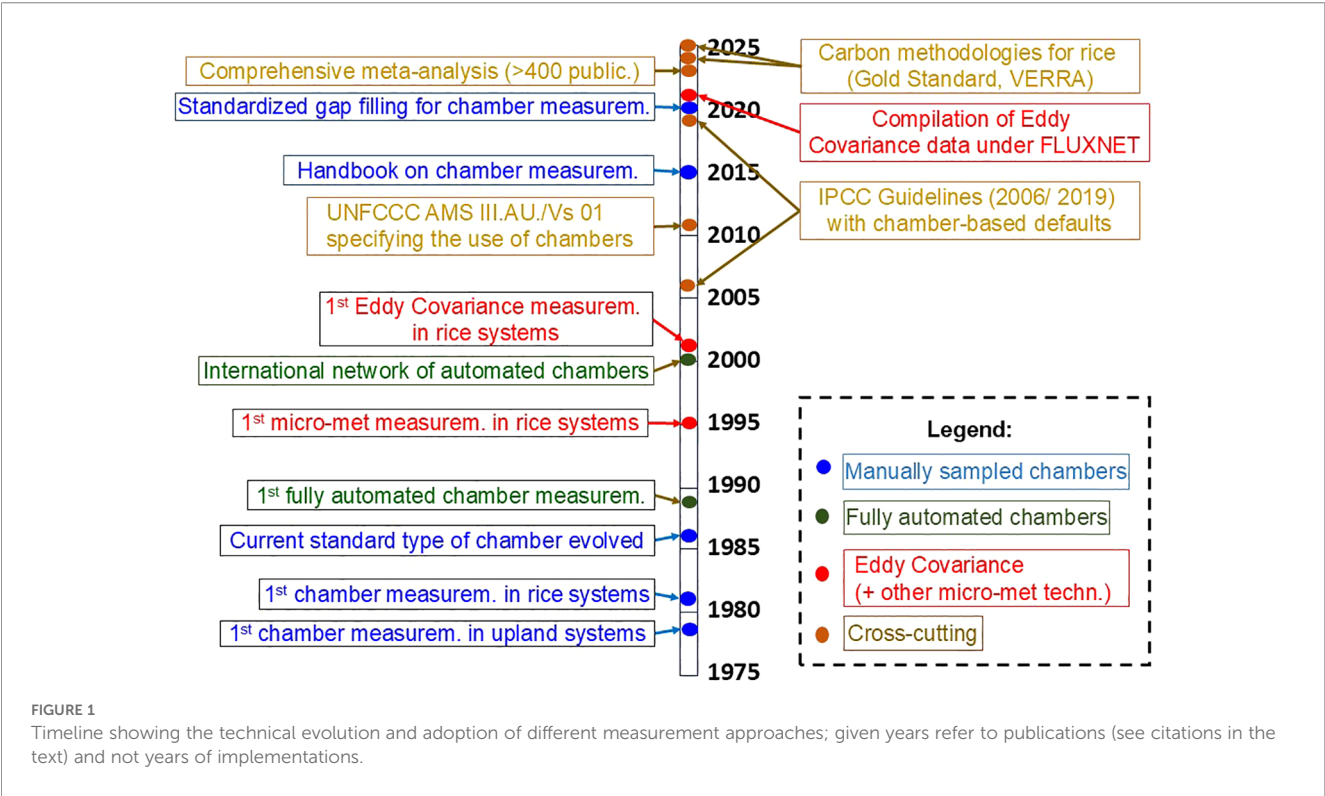
One of the obvious research questions for the newly established EC measurements in rice fields was the compatibility of the results with chamber measurements. Reba et al. (2020) found that the recorded similar fluxes obtained with both methods in the early and late stages of the rice season, but that they differed during mid-season, with the chambers consistently reporting the higher values. Likewise, Chaichana et al. (2018) recorded higher CH₄ fluxes obtained by chambers in the period from 30 to 90 Days after Transplanting. The reasons for these temporarily defined discrepancies could be related to peak emission sensitivity, typically occurring at mid-season. Given the complex interaction in determining the magnitude of CH₄ emissions, it seems plausible to assume larger spatial variations at periods of high emissions as compared to periods when emissions are generally at a low background level. EC measurements were also used to determine a strong El Niño effect, reducing annual CH₄ emissions of Thai rainfed rice (Setsungnern et al., 2025).

5 Discussion

Figure 1 shows a comprehensive timeline of the technical evolution and adoption of different measurement approaches based on the respective years of publications. In addition to the development of MSC, FAC and EC, we also highlight a few cross-cutting events, e.g. such as the publications of the IPCC Guidelines.

5.1 Decision-making selection framework

As a starting point for an integrative assessment of the scientific contributions obtained by different approaches, Figure 2 provides a schematic presentation of their different measurement frequencies and durations of the data points gathered with each approach. For comparative purposes, we developed an evaluation matrix to highlight insights and decision-making frameworks for method selection. The matrix comprises six criteria representing potential objectives for each approach alongside logistical requirements of field measurements in rice (Table 1). Moreover, we provide a visual presentation in the form of radial charts (Figures 3a, b) based on



scoring each approach for these criteria within a scale ranging from 1 (corresponding to a weakness of the respective approach) to 5 (corresponding to a strength. Given the prevalence of the manual sampling approach in research, this approach is regarded as the

reference approach (Figure 3a) that will also be used in Volume 2 of this paper for the comparative assessment with innovative technologies (Vo et al., 2026). The manual chamber clearly outcompetes the other two systems in terms of mobility for

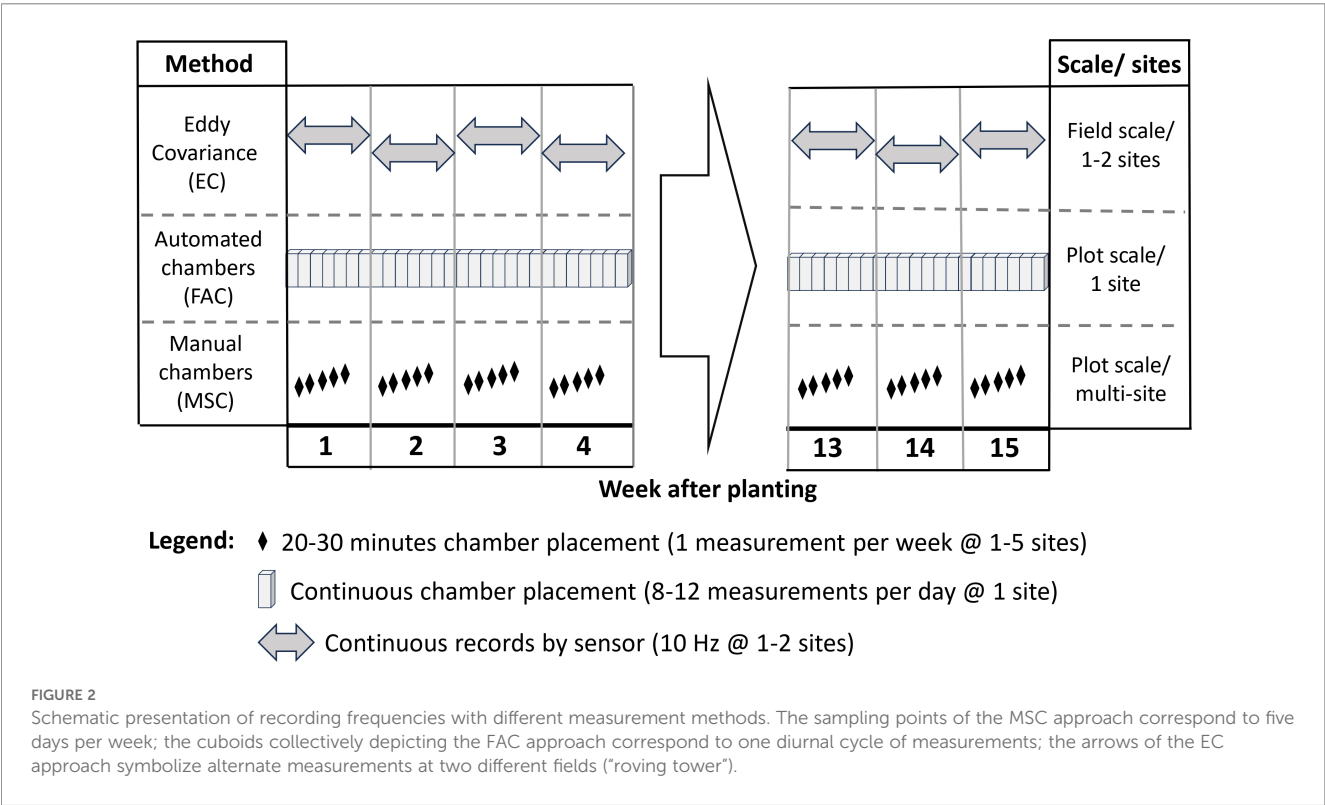


TABLE 1 Decision-making framework for the selection of measurement approaches for field measurements in rice.

Evaluation criteria	Manually sampled chambers	Fully automated chambers	Eddy covariance
1) Mobility for multi-site comparison	Very high mobility	No mobility in a given season	Very limited mobility and multiple treatment capacity due to required relocations of equipment and field size (min. 4 ha)
2) Velocity of screening	Suitable due to parallel measurements, but very labor-intensive	Suitable, but limited to comparing 3–4 treatments	
3) Recording of diurnal cycles	Low suitability due to laborious field operations	Very suitable due to automatic operations	Very suitable due to automatic operations
4) Accuracy of detection	Moderate accuracy of GC analysis	Moderate accuracy of GC analysis	Highly accurate sensors integrating at field scale
5) Affordability ¹⁾	Low costs as long as a GC lab is available	Affected by very high investment costs	Affected by very high investment costs
6) Simplicity of operation and data analysis	Very simple as long as a GC lab is operational	Highly technical know-how on operations required	Advanced know-how on data management required

¹⁾ Investment costs low: <10,000 USD; high: 30 - 50,000 USD; very high: > 50,000 USD.

multi-site comparison (#1), affordability (#5) and simplicity of operation and data analysis (#6), which can be seen as the main reasons for its popularity. As for the investment costs, however, it should be noted that we assumed access to a functional GC lab and excluded eventual costs for such infrastructural requirements of the manual chamber approach. Automated chambers perform strongly in recording diurnal cycles (#3) and receive a moderately high score for the velocity of field screening (Figure 3b) because of the limitation to 3–4 treatments that can be assessed in parallel. In all other criteria, automated chambers received only low scores. Finally, the Eddy Covariance system (Figure 3b) received the highest score for the recording of diurnal cycles as well as the accuracy of detection – as the only approach covering the field instead of the plot scale – received the highest score, whereas all other criteria were lowly scored.

5.2 Scientific evaluation framework

Chamber-based techniques are the most widely used approach for GHG emissions from rice fields, providing the foundation of the state of knowledge on GHG emissions in rice as affected by environmental and agronomic factors. This pivotal role will most likely also remain in the near future, as manually sampled chambers are the established standard method applied for national GHG inventories of most rice-growing countries and are also defined as the standard measurement approach in mitigation projects aiming at the voluntary carbon market. So far, this measurement approach has facilitated the empirical foundation for the various global estimates of the CH₄ source strength of rice production that are elucidated in the review by Qian et al. (2023). However, this approach has also received considerable criticism. The arguments raised against the MSC

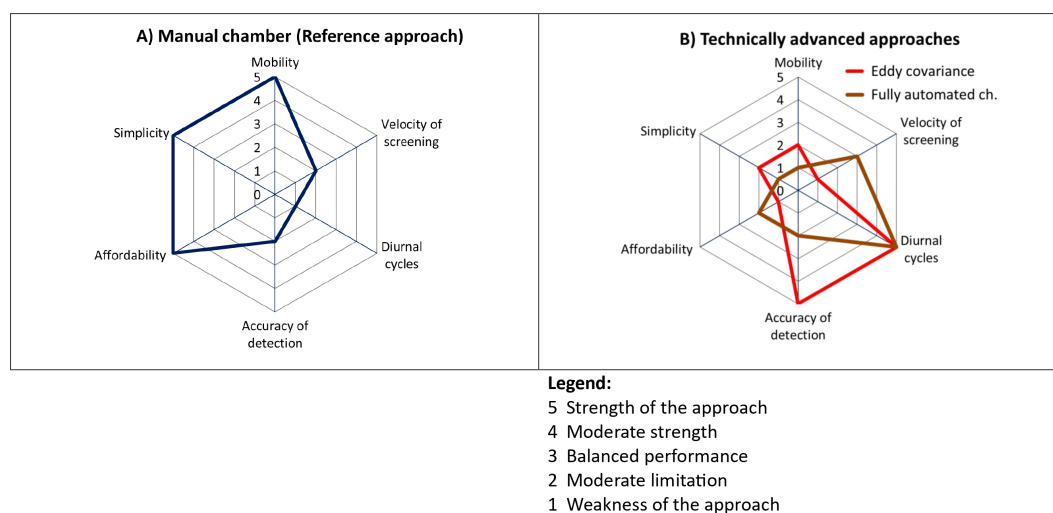


FIGURE 3

(A, B) Schematic presentation of strengths and weaknesses of all approaches according to six selection criteria for the specifics of GHG measurements in rice (see Table 1 for individual assessments).

approach are two-pronged, namely (i) an alteration of the micro-climate by excluding turbulence (Beauchamp et al., 1978; Mosier, 1990) and (ii) the obvious data gaps in terms of coverage of the variations over space and time that were elaborated for different modifications of the MSC by Sander and Wassmann (2014).

Regarding the argument about the altered micro-climate, this caveat is generic to all chamber approaches and also applies to the FAC approach (Table 2). However, many research questions inherently require comparative field testing of different treatments that have to be replicated for assessing statistical significance. The EC technique – as well as other micro-meteorological approaches – has other limitations (see below), so the critique is somewhat open-ended as there is no real alternative available at this point.

As for the inherent gaps in sampling points, it should be noted that the question of gap filling of chamber measurements was recently addressed through a standardization of procedures (Dorich et al., 2020) that largely correspond to the gap filling procedures previously developed for EC measurements (Falge et al., 2001). Looking at the entirety of studies, it seems justified to say that MSC was shown as a robust and versatile approach with sufficient accuracy to facilitate the participation of numerous research groups in the data generation for multi-site measurements, as well as comparative studies on crop management effects.

As compared to MSC, the FAC had a much smaller number of applications across different countries. This can be attributed to causes relating to both high investment costs and complex operations on one hand, as well as the inherent limitation of the FAC approach in multi-location studies on the other hand. In Volume 2 of this paper, we will present new laser-based analyzers allowing direct measurements in a field set-up that is very similar to the FAC systems. As these mobile systems require operational inputs to establish the chamber and valve system at each given site, however, they can best be described as semi-automatic to clarify the distinction from FAC systems that independently operate throughout the entire season. Irrespective of

the low number of established FAC systems, these studies yielded important information to substantiate the database on (i) diurnal and seasonal cycles and (ii) a process-based understanding of the impacts of different crop management practices, which are vital for parametrization and validation of process-based models (Kraus et al., 2015). It is understood, though, that these advantages come at the expense of the immobility of the measurement system, which will exclude this approach from various applications, e.g., obtaining region-specific emission factors. As another critical factor, the generic arguments pointing at the altered microclimate in the chamber headspace are even amplified as the chambers will remain at identical sites throughout the vegetation period.

Finally, the EC system is shown in Table 2 with prominent scientific contributions, but also apparent limitations for various research questions. While its integration up to the field scale provides more reliable data by excluding plot-scale variations, this approach entails fundamental limitations for comparative measurements of multiple sites and different treatments. Thus, the contributions obtained by EC measurements represent vital data sources whenever highly dynamic and spatially integrated data are needed, which may become especially valuable in the future for ground-truthing satellite-based flux measurements in combination with advanced computing technologies as ML.

5.3 Challenges in detecting very low flux rates of N₂O

The N₂O emission rates in rice fields are generally very low and many studies have reported near-zero or even negative levels (Linquist et al., 2012). Moreover, the events with sizable N₂O emissions are often confined to sporadic events over the cropping season, e.g. after fertilization or water draw-down. As for the MSC approach, the detection limit of conventional gas chromatography

TABLE 2 Scientific evaluation framework based on key findings and limitations associated with different measurement approaches in rice fields.

Evaluation criteria	Manually sampled chambers	Fully automated chambers	Eddy covariance
Method-specific key findings on GHG dynamics	- Baseline data on emission factors in different regions and seasons - Enormous wealth of available emission data allowing statistically meaningful analysis at the meta-level - Standard method for Tier 2 approach in GHG inventories and carbon projects	- High temporal resolution data on diurnal and seasonal cycles - Coherent data for comparison of crop management practices - Provides database for identifying optimal diurnal sampling times (0600, 1200, 1800 h) for representative manual sampling	- Integrating across spatial variations at field scale and thus, avoiding inherent shortcoming of chamber measurements - High resolution data on diurnal cycles - Linking emissions dynamics to ecosystem functioning
Limitations and controversies in current measurement practices	- Spatial variation may not be compensated by 3–4 replicates - Changing the micro-climate within the canopy by excluding turbulence - Uncertainty on how to evaluate non-linear concentration increases		- Required 100-m fetch limits its use in smallholder contexts and for comparative crop management assessments - Gap filling is frequently needed for emission data at night-time and during seasons with unstable wind conditions - Uncertainties in partitioning flux components and energy balance closure.
Specific objectives and envisaged improvements	- Mainly used to compare treatments - High potential for integrating improves analytics, e.g. laser-based trace gas analyzer (see Volume 2)		- Mainly used to estimate absolute emissions on a large scale - Great potential for networking among stations in combination with RS

may inhibit quantifying very low increments near background atmospheric concentration levels (~ 330 ppb N_2O). In a lab protocol published by Kahmark and Robertson (2023), the coefficient of variation (CV) should be less than 3%, i.e. this CV of N_2O would be around 10 ppb (3% of 330 ppb = 9.8 ppb) sensitivity or lowest detectable limit. While many field studies reported incidents of such low concentration increments, their leverage on the seasonally cumulated N_2O is typically very small as long as these low-emission episodes come up in succession with periods of moderate or higher emission rates within a given cropping season. In turn, it could be argued to what extent the precise detection of such small rates up to several digits will really be instrumental in the overall picture of N_2O emissions.

The analytical detection for GHG emissions (N_2O and CH_4 alike) can be improved through advanced technologies, namely laser-based analytics, which is discussed in detail in Volume 2 of this paper (Vo et al., 2026). As an alternative avenue with low investment costs, researchers could consider modifications in the field measurements to enhance the concentration increments and thus, make the quantification of a low emission rate much easier. One option would be extending the sampling intervals, but that would only be possible up to the maximum placement time of around 30 min that is considered a threshold to avoid plant damage. In turn, the sample extraction could reach a 30-minute interval, but that would come at the expense of having only two sampling points per chamber placement. The second technical option is reducing the height of the chambers, but that cannot be reconciled with capturing CH_4 that largely emits through the plant aerenchyma (Butterbach-Bahl et al., 1997). However, the majority of mechanistic studies on N_2O indicate that this gas is primarily emitted from the soil surface rather than transported through the plant (Timilsina et al., 2020). Although not mentioned in any of the incumbent guidelines or methodologies, using shallow chambers placed between plant rows for longer sampling intervals could be seen as a potential alternative for quantifying N_2O emerging from the soil. Even if this approach neglects the plant-mediated N_2O which may or may not contribute to the total flux, this caveat has to be seen against the option of using high static chambers that are not suitable to detect either transport pathway with conventional GC analytics. As for the future, targeted field campaigns using high-resolution analytics could provide more clues on the best strategy to deploy specifically designed chambers for assessing N_2O emissions.

Regarding the irregular occurrence of spikes in N_2O emissions, these seasonal patterns could, in principle, be covered by FAC and EC systems. While most FAC systems previously deployed in rice fields consisted of high chambers optimized for CH_4 emissions, shallow chambers were also integrated in an automated system specifically designed for recording N_2O and nitric oxide emissions from rice fields (Gaihre et al., 2015). Likewise, the scope for measuring N_2O emissions from rice fields using the EC technique is promising, but still limited (Li et al., 2023). Installing the sensors on a high tower enhances the spatial footprint and thus could level the background variability of N_2O fluxes at the landscape scale (Xie et al., 2022).

6 Conclusion

In brief, MSC pioneered GHG measurements more than four decades ago and will almost certainly remain as the method of choice for the bulk of field studies in the near future. As for the alternative approaches presented here, both FAC and EC entail major technical and financial obstacles that will impede a wider adoption for routine GHG inventories or research projects on mitigation strategies. However, these high-tech approaches have played important roles in addressing specific research questions that can hardly be dealt with by MSC. This scientific niche will determine their applications – though limited in number – for providing new insights into the mechanistic drivers of GHG emissions.

Irrespective of the measurement approach, any projection of future measurement approaches will also have to consider ML and Artificial Intelligence (AI) technologies as real gamechangers for making field data more reliable and affordable. At this point, ML and AI technologies may still need the provision of a critical mass of data for development and validation as stand-alone approaches and are generally seen as being part of future Tier 3 approaches, e.g. as reflected in the new VERRA rice methodology (VERRA, 2025). On the other hand, they may also become very useful tools for gap-filling and extrapolations under a Tier 2 framework, which will further be elucidated in Volume 2. Moreover, Volume 2 explores the next step in technical innovations, which we anticipate in the form of an upgraded MSC approach using laser-based technologies. Thus, the scope of this 2-volume paper ranges from a synthesis of previous scientific findings, distinguishing between their origin through different measurement approaches, to outlining cutting-edge field set-ups of the future. This multilayered narrative provides the rationale for publishing two volumes under different paper categories of this journal, namely one review paper (Volume 1), leading to one method paper (Volume 2).

Further research and development initiatives may consider increasing data accessibility in GHG emissions through more targeted and coherent networking efforts. The FLUXNET network of the EC community could serve as a role model for compiling results from different groups in coherent formats of global data sets with open access for interested researchers. This type of networking will be the key to advancing global efforts in climate action and leveraging the contribution of rice to the transition to net zero for major rice-growing countries. As detailed in Volume 2, the convergence of these inter-regional datasets with new technologies—specifically ML and RS on proxies like water levels—will be critical for refining global GHG estimates. While we acknowledge that direct satellite-based CH_4 detection remains a distant prospect due to current sensor limitations, the role of field measurements will only intensify. They will serve as the indispensable ground-truthing and calibration backbone, providing a convincing counterargument to the notion that field-based evidence may soon be rendered obsolete by new technologies.

Author contributions

TBTV: Methodology, Writing – original draft, Writing – review & editing. RW: Conceptualization, Methodology, Visualization, Writing – original draft, Writing – review & editing. BOS: Writing – review & editing. AMR: Supervision, Writing – review & editing.

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conventional methods and the current submission focuses on the technological development of greenhouse gas quantification in rice systems. Both manuscripts follow related methodologies but differ in scope, objectives, and contributions.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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